

Negative Exponents with Variable Bases*Question Bank - Additional 3.T2*

Show your work. Circle your final answer.

For questions 1 through 6, write an equivalent expression using only positive exponents. Assume no variable equals 0.

1. b^{-8}

2. $\frac{1}{m^{-4}}$

3. $9x^{-5}$

4. a^4c^{-3}

5. $\frac{p^2}{q^{-6}}$

6. $\frac{4x^{-3}y}{z^{-5}}$

For questions 7 and 8, apply the laws of exponents. Write each answer using only positive exponents.

7. Write an equivalent expression.

a. $y^2 \cdot y^{-9}$

b. $(2m^{-4})^3$

c. $(3x^{-5}y^3)(4x^8y^{-7})$

8. Write an equivalent expression.

a. $\frac{k^3}{k^{10}}$

b. $\frac{20a^2b^{-3}}{5a^6b}$

c. $\left(\frac{x^2}{3y^{-2}}\right)^3$

Working backwards.

9. Find the value of n that makes each statement true.

a. $a^{-3} \cdot a^n = a^4$

b. $\frac{x^n}{x^5} = \frac{1}{x^2}$

c. $(y^n)^4 = \frac{1}{y^{20}}$

10. Let $x = 4$.

a. Find the value of x^3 and the value of x^{-3} .

b. Find the value of $-x^3$. Is it equal to x^{-3} ? Explain.

11. Which expression is equivalent to $2y^{-4}$?

- ☐ A. $\frac{2}{y^4}$
- ☐ B. $\frac{1}{2y^4}$
- ☐ C. $-2y^4$
- ☐ D. $\frac{1}{16y^4}$

12. Select ALL of the expressions that are equivalent to $\frac{1}{m^8}$.

- ☐ A. m^{-8}
- ☐ B. $m^3 \cdot m^{-11}$
- ☐ C. $\frac{m^2}{m^{10}}$
- ☐ D. $(m^{-4})^2$
- ☐ E. $-m^8$
- ☐ F. $(m^{-2})^{-4}$